



MINING DEALER BEST PRACTICE

Auxiliary Drive Shaft Bolt Tightening Tool for Engine Auxiliary Drive Shaft C175

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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July 2024
4.03-240731-1

1.0 Introduction

A new tool has been designed for tightening the rear PTO drive shaft bolts on C175 engines.

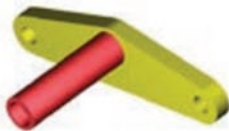
2.0 Best Practice Description

This Best Practice covers an upgraded design to the standard CAT tool P/N 478-1934. The standard Cat tool allows 9 of the 12 rear PTO bolts on the C175 engine to be tightened before the tool is repositioned, and due to cylindrical geometry of the bolt contact area, creates a potential for contaminate generation with constant use. The new design is shaped to allow 11 of the 12 rear PTO bolts to be tightened before repositioning the tool and utilizes a grooved bolt contact area that minimizes the risk of generating contamination.

PTO Shaft Stop Tool Assembly

Model Usage: All C175s

Warranty: Caterpillar® Six Months



- Attaches to the rear housing assembly to captivate the PTO shaft
- Prevents the PTO assembly from rotating when assembling the shaft onto the auxiliary drive gear
- Simplifies the assembly process

Specifications

- Dimensions: 92 x 48 x 102 mm (3.6 x 1.9 x 4.0in)
- Weight: 1 kg (2.2 lb)

Part No.	Description
478-1934	PTO Shaft Stop Tool Assembly

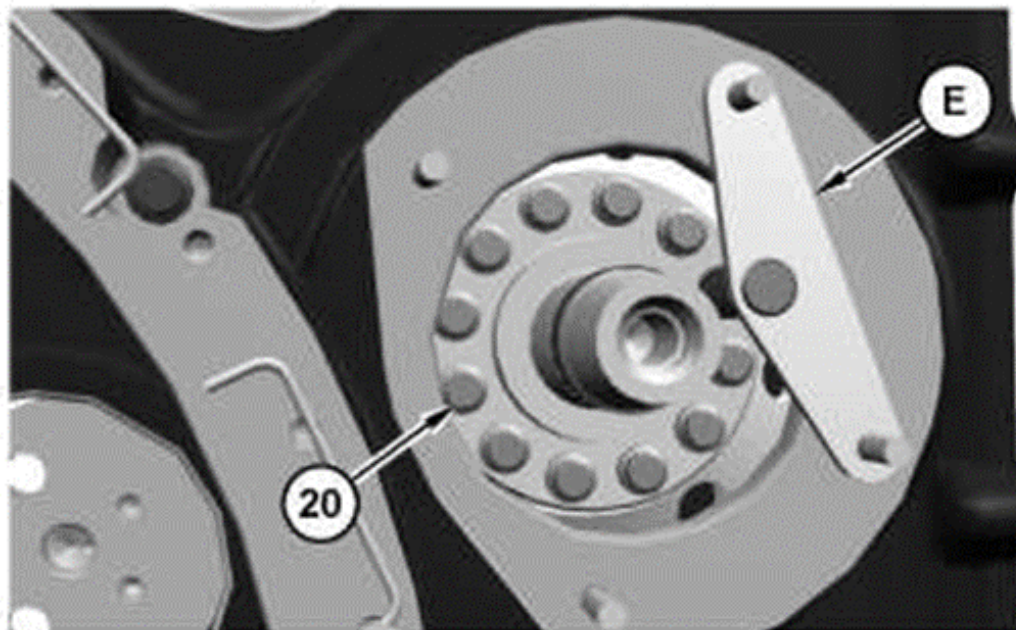


Figure 1: Tool N/P 478-1934

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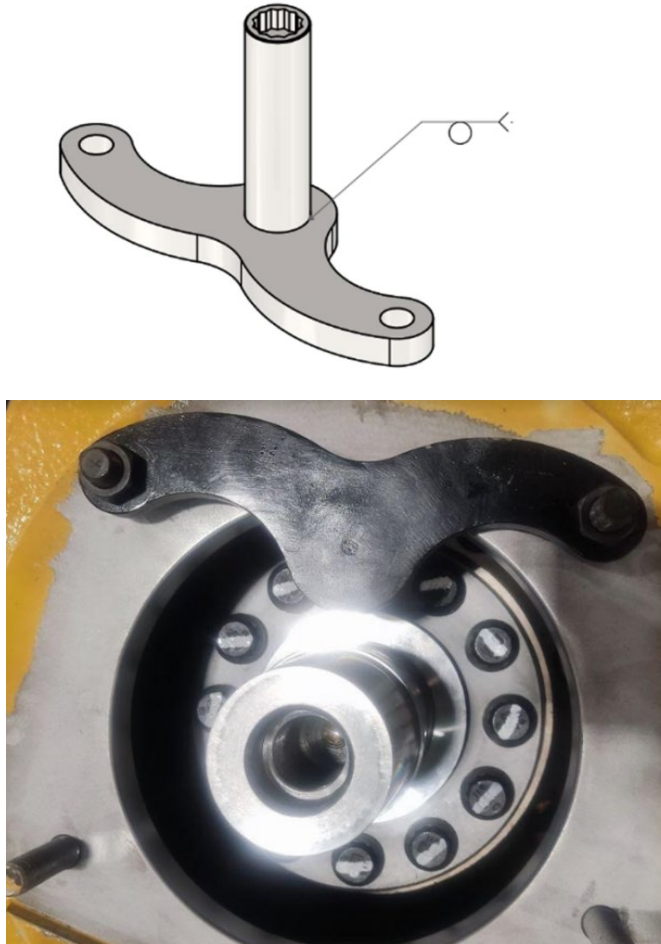


Figure 2: Modified tool design for tightening the rear PTO drive shaft bolts of C175 engines.

3.0 Implementation Steps

The use of the new tool is similar to the CAT tool (478-1934) it is placed in the rear housing and allows to lock one bolt of the rear PTO drive shaft of the C175 engines and tighten 11 bolts to the specified torque, then turn to lock another bolt and tighten the last bolt.



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4.0 Benefits

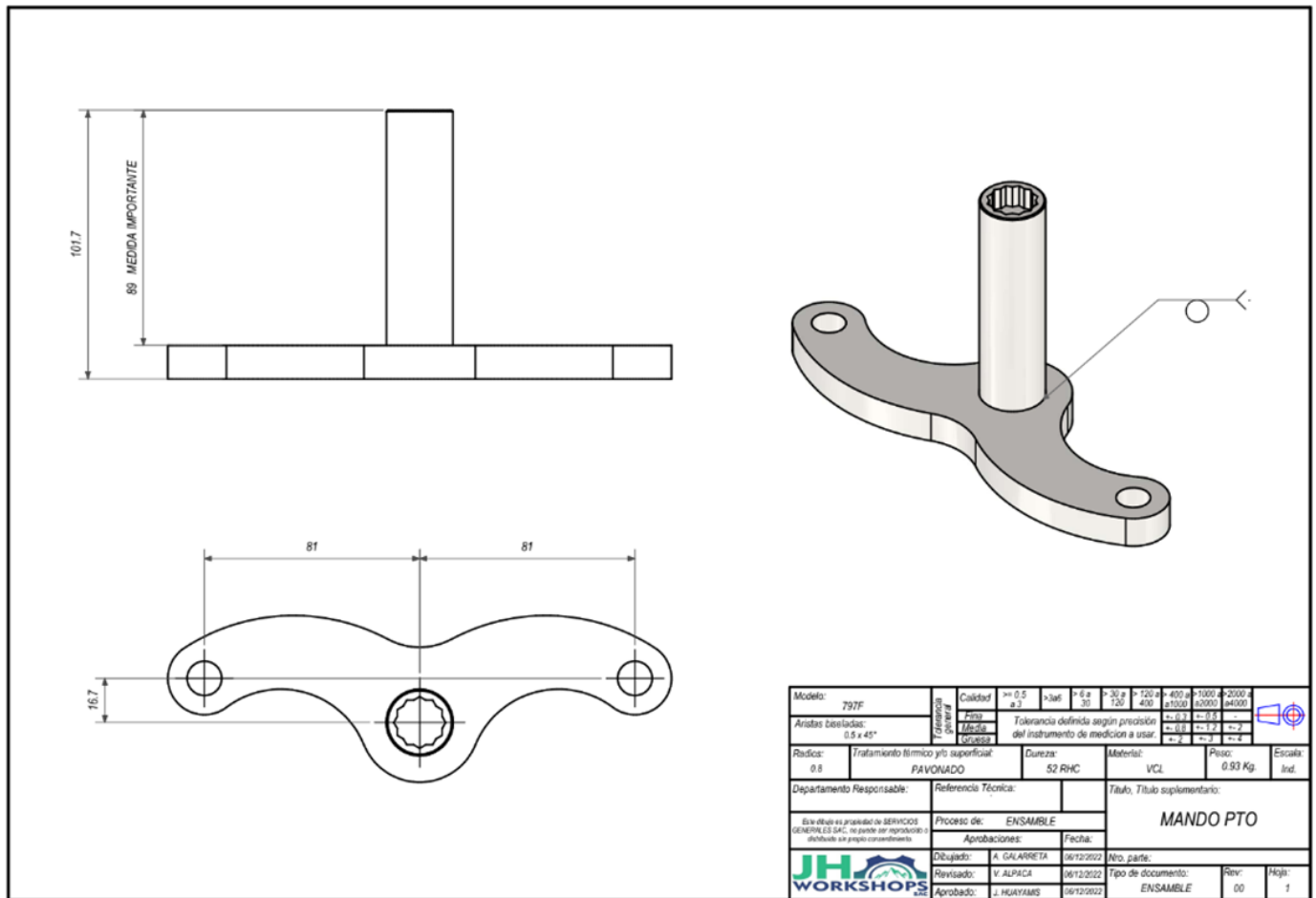
- It allows to adjust in the first stage up to 11 bolts of the 12 that hold the shaft of the rear PTO drive of C175 engine, then just finish with the bolt that was used as a lock.
- In addition, the knurled shape of the tool does not generate any contaminants when using the tool for bolt tightening.

5.0 Resources Required

- Fabrication of the new tool according to drawing.
- Manufacturing costs ~ 320,00 USD

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6.0 Supporting Attachments / References



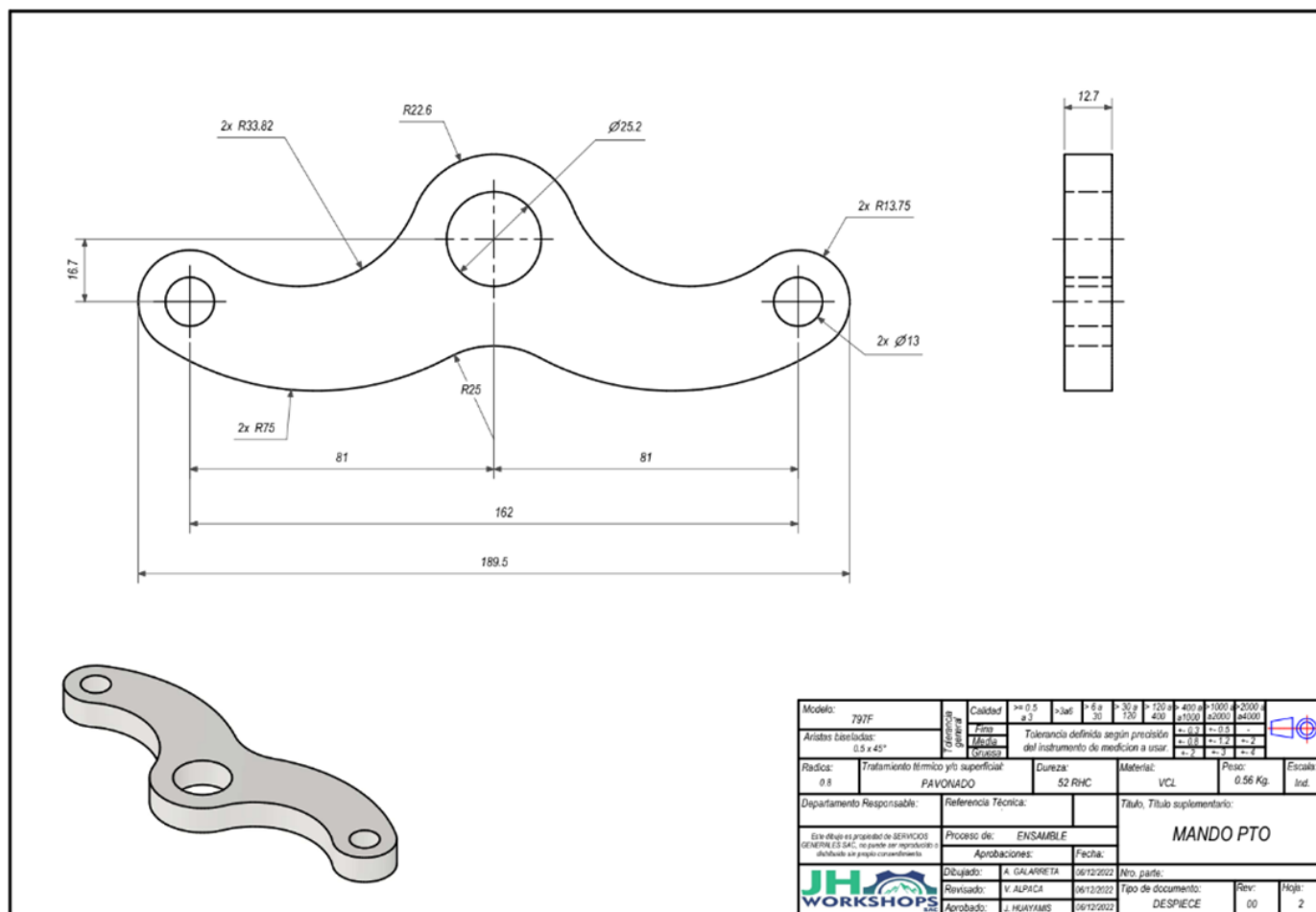
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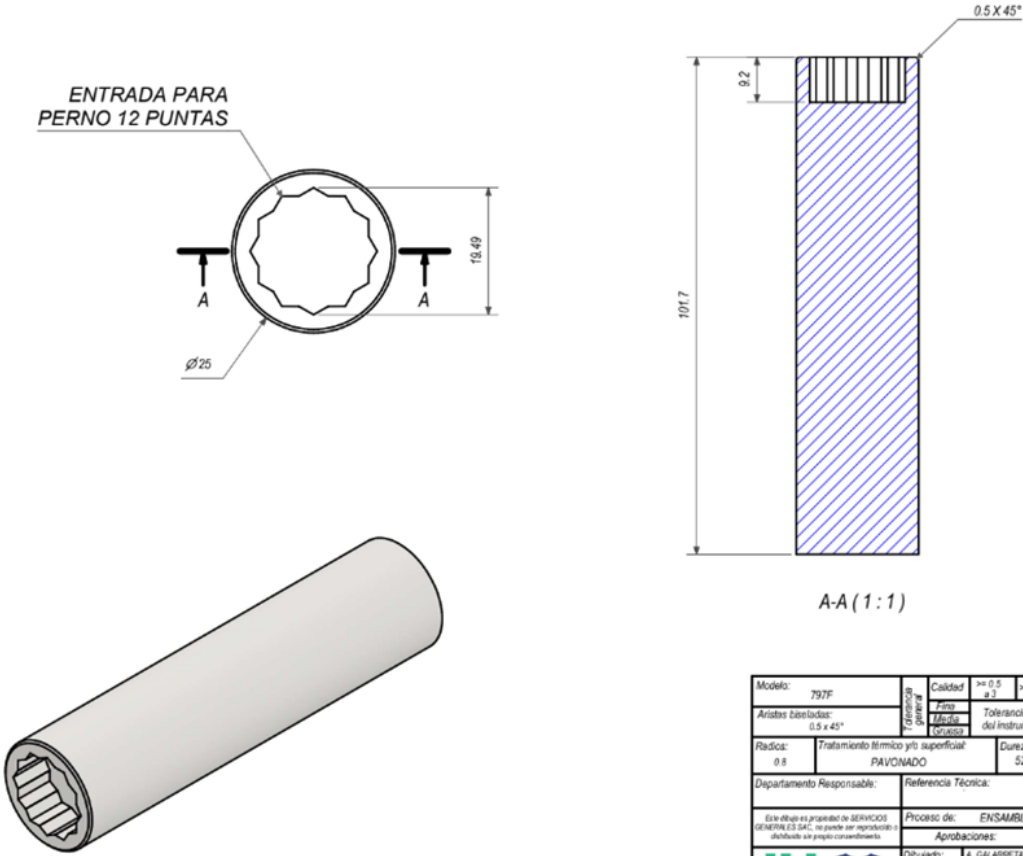
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None

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